

Pankaj Thakur

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Gwalior, India

PROFESSIONAL SUMMARY

AI & Full-Stack Developer pursuing B.Tech in AI & Data Science at MITS Gwalior (2027). Specialises in **Generative AI**, **Retrieval-Augmented Generation (RAG)**, and **multi-agent LLM systems** using LangChain, LangGraph, and Groq. Shipped three production-shipped projects: a 5-agent AI research dashboard with real-time SSE streaming, a 7-stage RAG document chatbot with persistent ChromaDB vector storage, and a full-stack parking SaaS with 3-role RBAC and Razorpay integration. **Top 7.3% nationally** in GATE CSE 2026. Seeking AI Engineer / SDE internship to build and scale real-world intelligent systems.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior

Sep 2023 – Aug 2027

B.Tech — Artificial Intelligence & Data Science | **CGPA: 8.3/10**

Relevant Coursework: Data Structures & Algorithms, DBMS, OOP, Operating Systems, Machine Learning, NLP, Computer Networks

TECHNICAL SKILLS

Languages	Python, C++, JavaScript (ES6+), SQL, HTML5, CSS3
Frontend	React.js, Vite, Tailwind CSS, React Router DOM, React Markdown, Responsive Web Design
Backend	Node.js, Express.js, FastAPI, REST API Design, JWT Authentication, RBAC, Middleware Validation
Databases	MongoDB, MongoDB Atlas, MySQL, ChromaDB (Vector Store), SQLite
AI / GenAI	LangChain, LangGraph, RAG, HuggingFace Embeddings, Prompt Engineering, Agentic AI, Multi-Agent Systems, NLP, LLM Apps, Generative AI, Mistral AI, Groq
Tools	Git, GitHub, Docker, Docker Compose, Nginx, Postman, Tavily API, Razorpay, Google OAuth, Server-Sent Events (SSE)
Deployment	Vercel, Render, MongoDB Atlas, Docker Compose

PROJECTS

OrchestrAI [\[GitHub\]](#)

May 2026

Tech: FastAPI | LangGraph | LangChain | Groq | React.js | Docker | SQLite | SSE

- **Architected** a full-stack **multi-agent AI research platform** with **5 autonomous agents** (Search, Reader, Writer, Critic, Final-Report) via LangGraph + Groq; containerised via Docker Compose + Nginx with SQLite run history, light/dark localStorage theme, and simulation mode for API-free demos.
- **Implemented** a real-time SSE pipeline with milestone-first execution logs and collapsible raw event panels; implemented **durable final_report state** so generated reports and critic feedback survive React re-renders without polling.
- **Built** copy/download bundling report + critic review, React Markdown rendering with a dedicated critic section, and scroll-isolated panels to prevent page-jump UX during multi-stage sequential pipeline execution.

DocMind AI [\[GitHub\]](#)

Apr 2026

Tech: Python | LangChain | ChromaDB | HuggingFace Embeddings | Mistral AI | Streamlit | RAG

- **Developed** an AI-powered PDF/TXT chatbot implementing a complete **7-stage RAG pipeline** (load → chunk → embed → store → retrieve → augment → generate) with LangChain, HuggingFace all-MiniLM-L6-v2 embeddings, and **persistent ChromaDB vector storage** eliminating re-embedding on repeated queries.
- **Replaced** keyword lookup with **semantic similarity retrieval** via ChromaDB; verified the full RAG flow through terminal debug logs confirming retrieved chunk count before each Mistral Large LLM call.
- **Grounded** LLM answers in retrieved document context to reduce hallucinations; implemented **5 personality modes** (Professional, Teacher, Motivational, Concise, Friendly) adjusting response tone per selection across PDF and TXT corpora.

SmartPark [\[Live\]](#) [\[GitHub\]](#)

Mar 2026

Tech: React.js | Node.js | Express.js | MongoDB Atlas | JWT | Razorpay | Vercel | Render

- **Architected** a live production parking SaaS with **3-role RBAC** (Driver/Owner/Admin) and reservation-aware occupancy ($Available = Total - Active Reserved$), excluding cancelled, failed, expired, and refunded bookings to prevent double-booking.
- **Engineered 23 RESTful API endpoints** (auth, parking, booking, payment, owner, admin) with JWT, Helmet, rate limiting, XSS protection, and Mongo sanitisation; hosted frontend on Vercel, backend on Render, data on MongoDB Atlas.
- **Integrated Razorpay** (create-order → verify → webhook) and Google OAuth; built owner earnings analytics (successful bookings only) and admin monitoring dashboard across full lifecycle: pending → confirmed → active → fulfilled or cancelled.

ACHIEVEMENTS & CERTIFICATIONS

- **GATE CSE 2026** — AIR 14,658 / 200,000+ candidates (**Top 7.3% nationally**), Score 426; strong fundamentals across Algorithms, DBMS, OS, and Computer Networks.
- **LeetCode** — **120+ problems solved** (arrays, trees, graphs, DP, hashing, recursion) with consistent Medium/Hard practice.
- **Data Structures & Algorithms** | **PW Skills** — Algorithmic problem solving, time & space complexity analysis.
- **Backend Development (Node.js & Express.js)** | **Udemy** — REST APIs, server architecture, database design, JWT authentication.